EXAMINATION

Luleå University of Technology

Course: **D7024E**
Course name: **Distributed and Mobile Computing Systems**
Date: **2020-10-26**
Time: **9.00-13.00, online home exam**
Aid: **WWW (no interactive communication with others)**

Teacher on duty (complete telephone number): Olov Schelén, **0920 492237** **olov.schelen@ltu.se**	**Teacher on duty (complete telephone number):**
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Grade scale: **50% grade 3, 75% grade 4 and 90% grade 5**	Note that this grade is for the exam. The exam is supposed to be easy enough so that anyone that followed the course properly should pass it. The grade for the course (including all assessment criteria) may be different.
Total number of questions and score: **3 questions, total score 36**	
Other information: This is an individual home exam.	Note: No interactive cooperation is allowed. Information can be searched on the world wide web. Please provide references, when such information is used in you answers.

General instructions

Check that you have received all the tasks/questions. You shall formulate answers with your own words, but for parts of the answers you may cite sources on the web. In any case you must provide the references to your primary sources (e.g., URL).

After examination

Examination results are posted in Canvas within 20 workdays after the examination.

1. **Mixed question on distributed systems (11p)**

   These are multi-choice questions, with three alternatives. Please answer "true", "false" or “depends” to each of the following statements. If your answer is “depends” you must provide a short explanation to get credits. For “true” or “false”, no explanation is required, unless you want to explain what conditions are assumed. If you have identified a corner case and provide a good explanation, we can accept an answer that is different from the expected one. Each question is worth 0.5p.

   a. In a distributed system, nodes typically synchronize on critical regions by using semaphores.

   b. Nodes in a distributed system typically share basic resources such as RAM and clock.

   c. Inter-process communication can take place via shared memory or via message transfer.

   d. Linear horizontal scalability means that adding more nodes to a distributed system also adds total performance.

   e. Global distributed consensus is a typical example where high scalability is obtained.

   f. Marshaling (a k a serialization) provides a unique and irreversible mapping to a unique key.

   g. Kademlia routing is done in constant time, independently of the number of nodes.

   h. The number of k buckets in Kademlia routing depends on the size of the address space.

   i. Distributed Hash Tables (DHTs) typically stores immutable data objects.

   j. In traditional SQL databases the transaction log (or command log) is typically used locally to handle transaction processing and to support a RAM cache.

   k. In SQL databases there is a need for a transaction log (or command log) to handle and resolve deadlocks.

   l. Many blockchain systems are based on a globally synchronized and replicated log of the command history since the inception/instantiation of the system.

   m. Blockchain solutions typically replicate all data to all nodes that are part.

   n. Leader based global consensus solutions typically assume that there can be a small number of traitors.

o. Leader based global consensus is an efficient way of providing non-Byzantine consensus.

p. NAT/firewall traversal by TURN is efficient in terms of handling payload data.

q. Modern DNS hosting services can heart-beat with the servers providing a service and advertise only those that have a heartbeat.

r. A pubsub service is typically implemented as a scalable distributed system that shards and replicates channels or topics to subsets of servers.

s. Kafka relies on global consensus to keep track or responsibilities and leadership.

t. If liveness is provided, safety is automatically obtained, but not vice versa.

u. Raft is a good intermediate for handshaking between nodes and keep track of which nodes are alive.

v. Certificate authorities are trusted entities that certify the ownership of public keys.

Note: In the following questions 2 and 3, it is important to include sources (e.g., URLs) and that you explain the answer in your own words.

2. **Miscellaneous (12p)**
   a) What are the major properties of a distributed hash table? Include also typical properties of the data. (4p)
   b) In Map-Reduce (e.g., Hadoop), in which ways is the communication of data minimized? (3p)
   c) In Microservices, how is database separation/distribution typically made? What tradeoffs are made (i.e., which advantages vs disadvantages do we get) (3p)
   d) In PGP, why are symmetric keys used in addition to the public-private key pair? (2p)

3. **Blockchain (13p)**

   a) Explain the principles of proof of stake. (4p)
   b) What are the major challenges with **delegated** proof of stake (e.g., as used by EOS)? (3p)
   c) In Ethereum, can multiple smart contracts be executed concurrently? Explain. (6p)